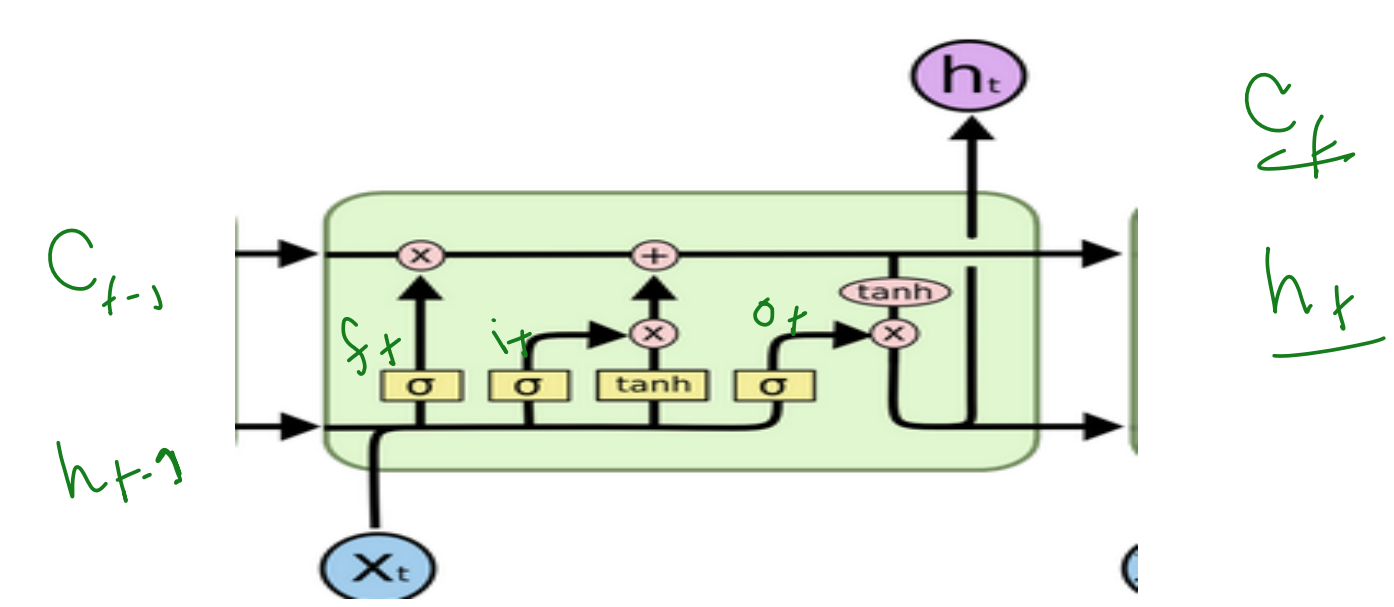




C_t, h_t

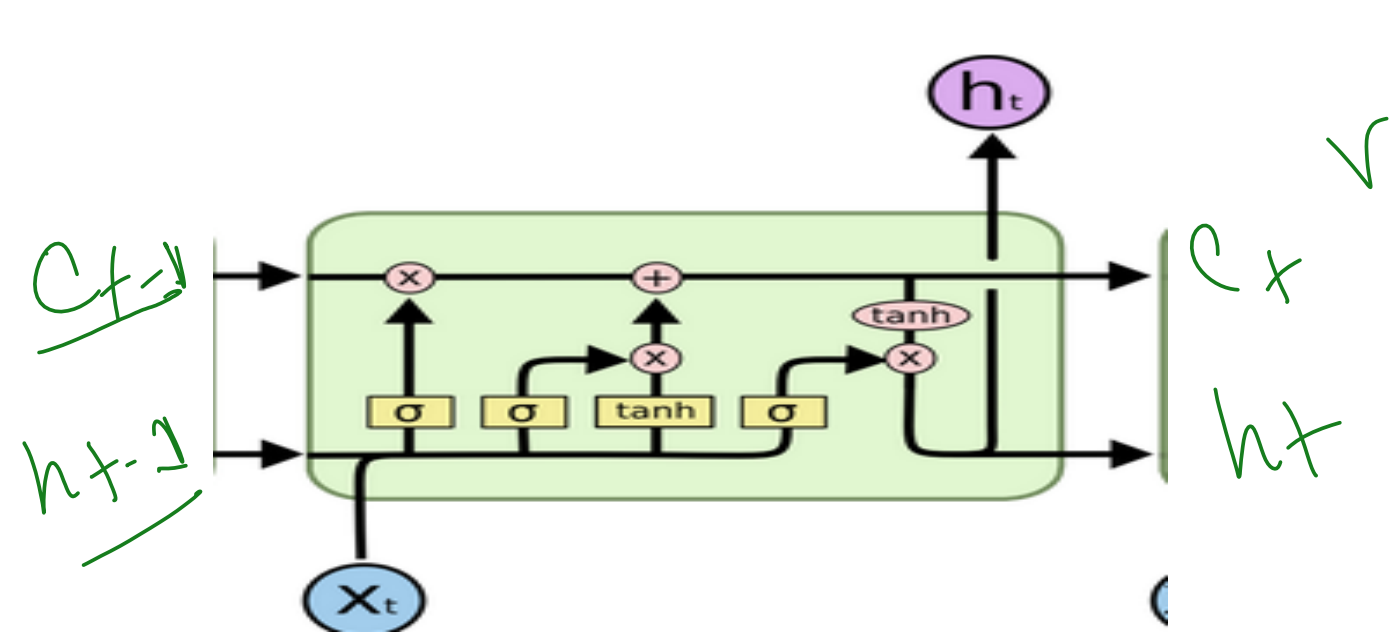
→ vectors → C_t, h_t dim 4096

Ex

$$C_t \Rightarrow \begin{bmatrix} 0.3 & 0.2 & 0.9 \end{bmatrix}$$

$$h_t \Rightarrow \begin{bmatrix} 0.1 & 0.7 & 0.4 \end{bmatrix}$$

What $x_t \rightarrow$ input



vectors → lion zebra horse

[Sentiment Analysis]

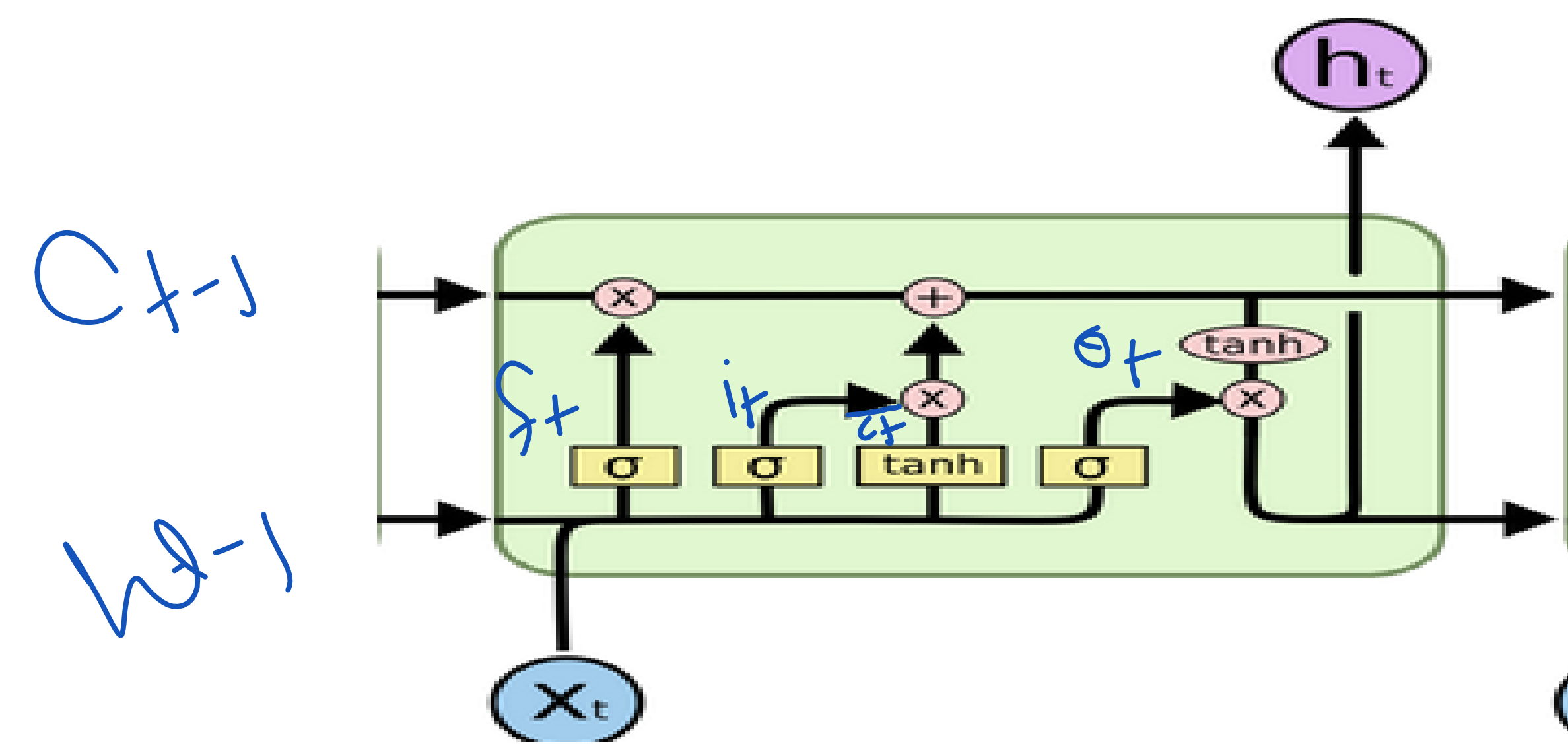
$t=1,2,3$

$$\begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}$$

lion zebra horse			0
zebra	lion	horse	1
horse	lion	zebra	1

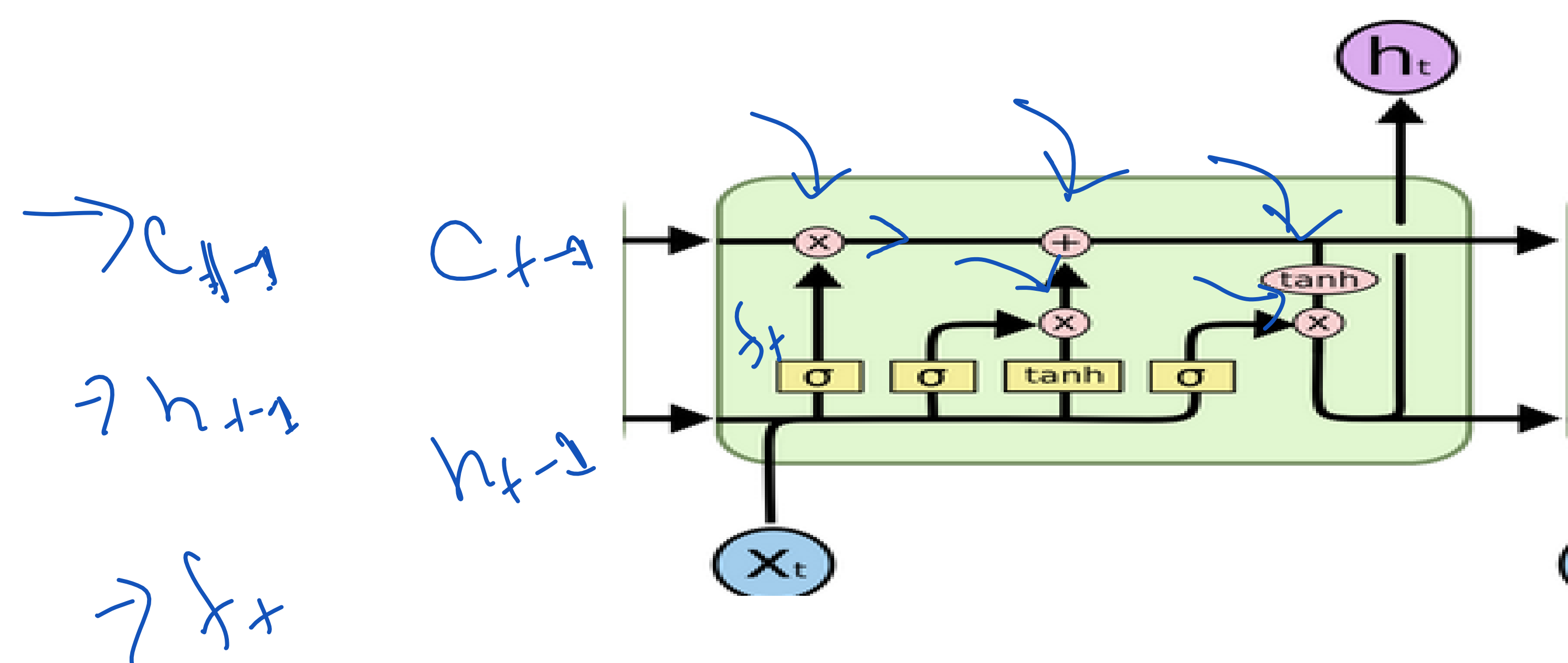
what are $f_t, i_t, \bar{C}_t, o_t$? → vectors

→ candidate cell state



$C_t, h_t \rightarrow$ dim same
 $s_t, i_t, \bar{C}_t, o_t \rightarrow \text{dim same}$

Pointwise operation:



$\otimes \rightarrow$ pointwise multiplication

$\oplus \rightarrow$ pointwise addition

— line

$\rightarrow i_1$
 $\rightarrow 0x$
 $\rightarrow \bar{C}_x$

$\text{tann} \rightarrow \text{point v.}$
 tann

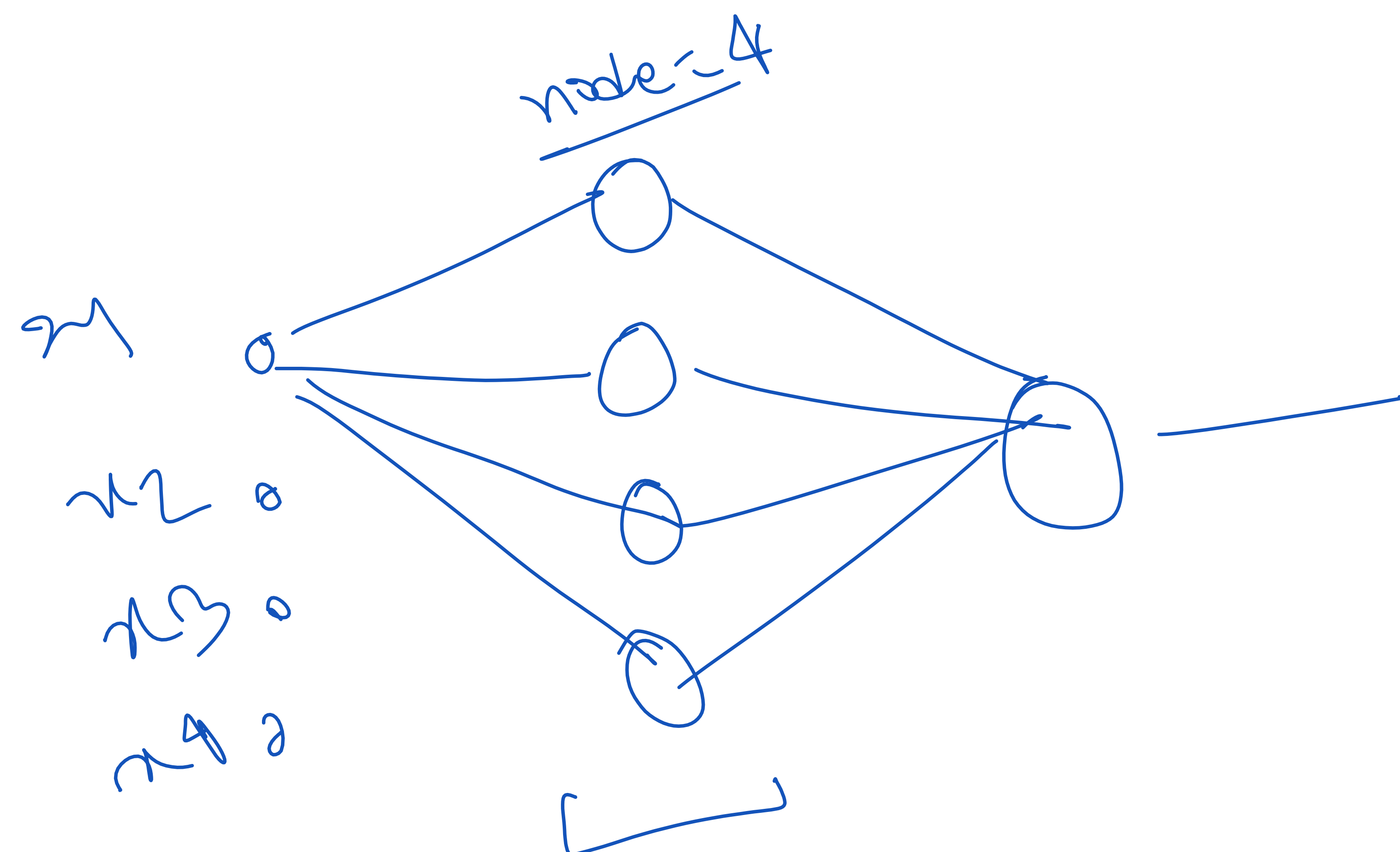
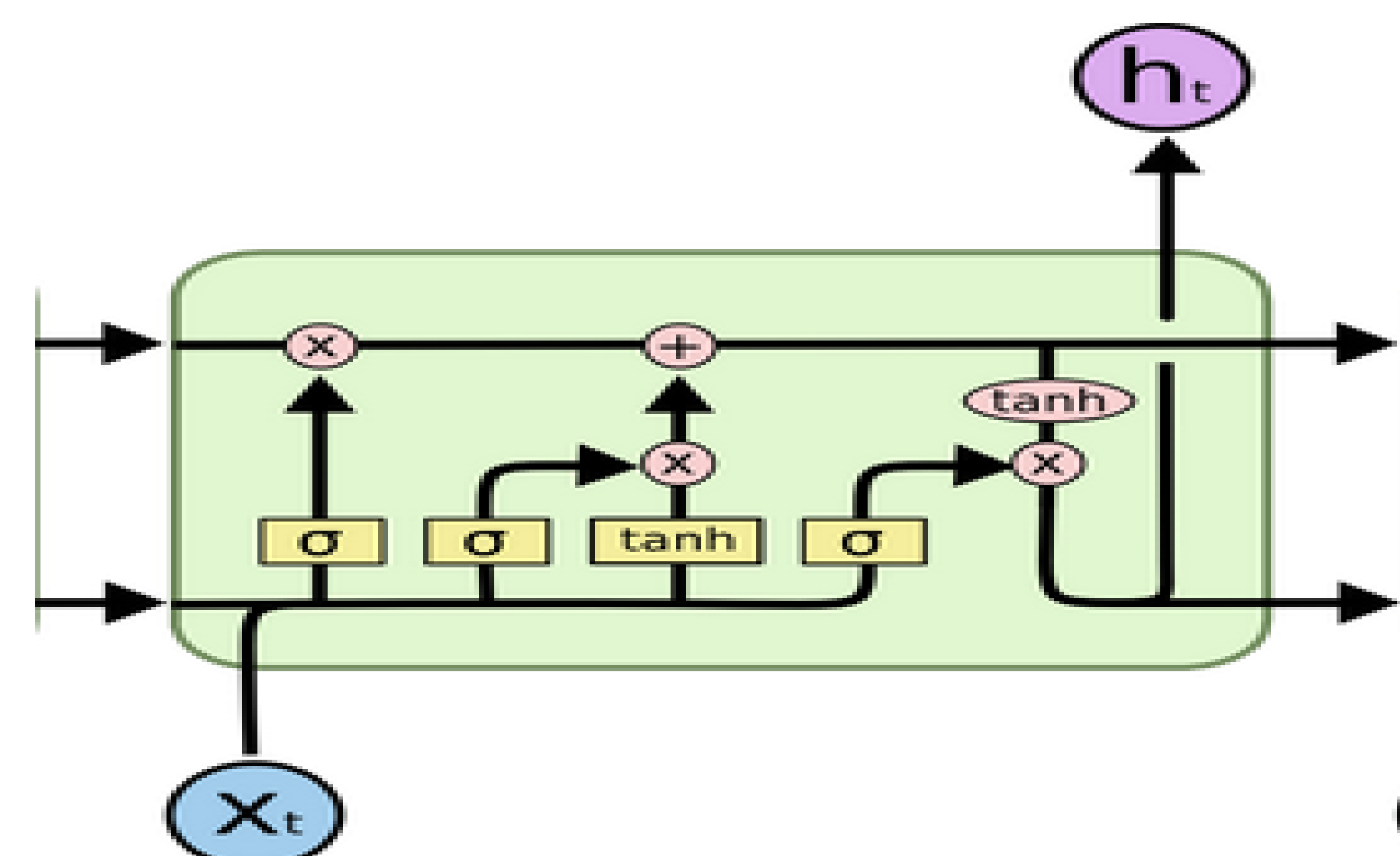


$C_{f-1} \oplus f_1 \Rightarrow [3 \ 8 \ 15] \rightarrow \text{vector}$

$\rightarrow$ point wise operation
 or
 dim matrix

$2^0 \ 2^1$
 $2^2 \ 2^3$

neural network layer



node = 4

→ 1 6

✓ ⑥ ⑥ ⑥

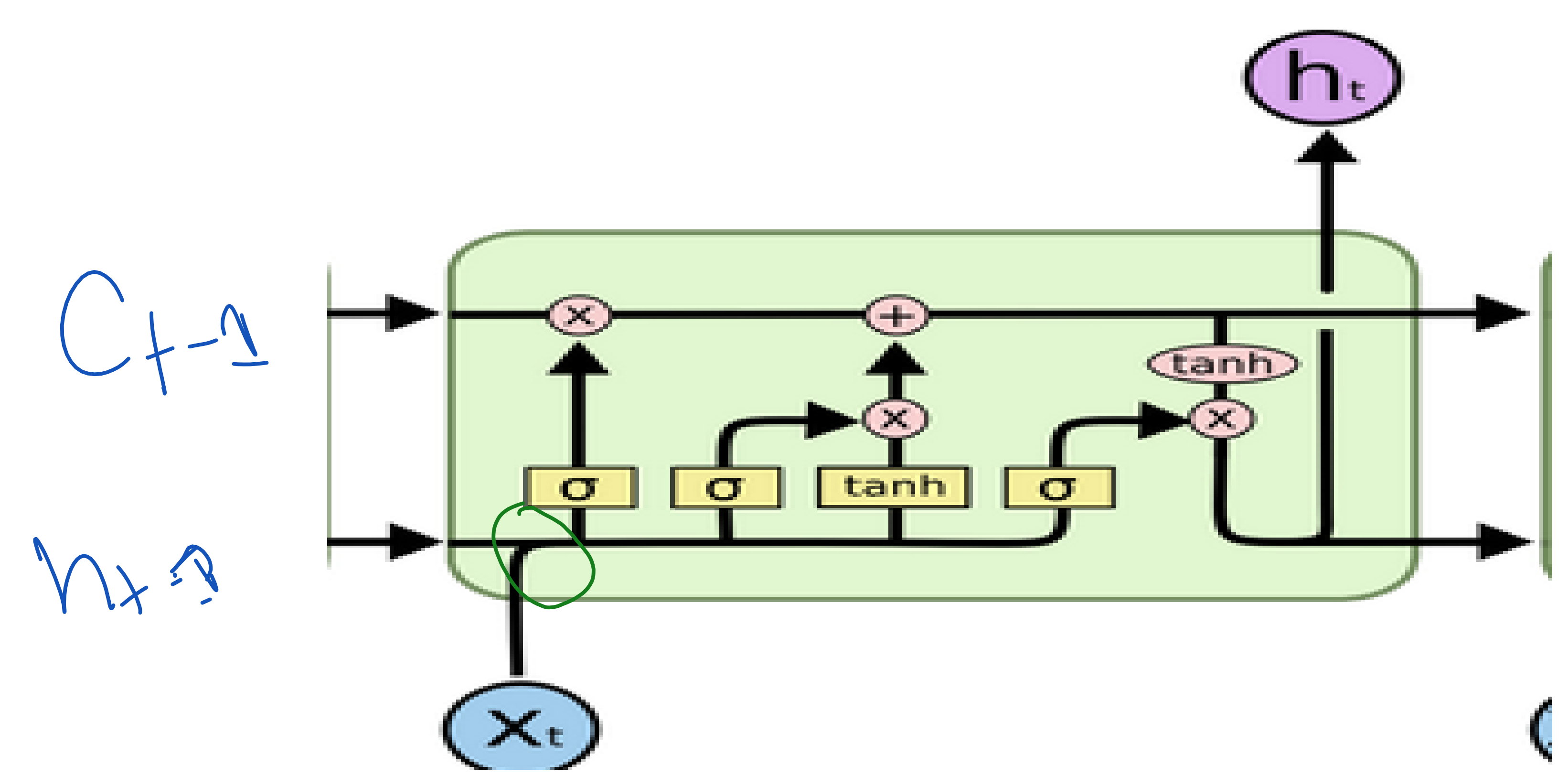
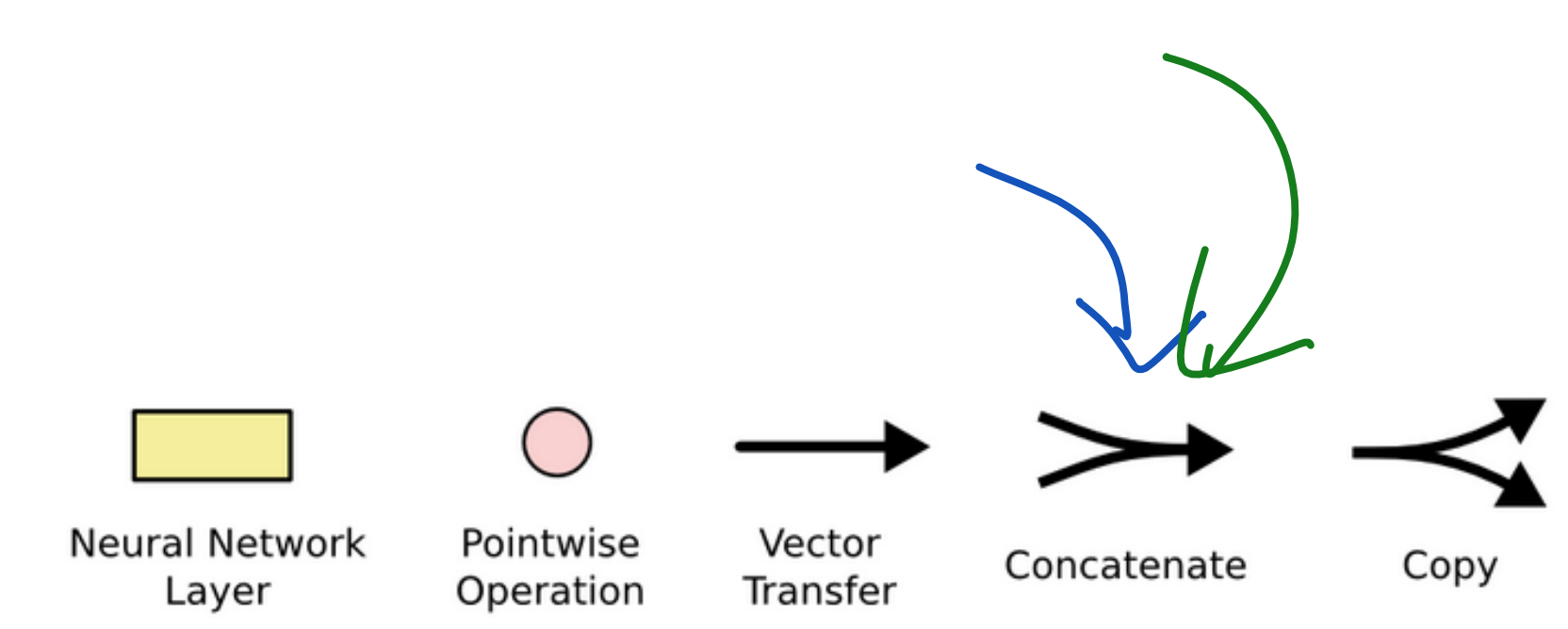
tanh

tanh

tanh

tanh

tanh



$h_{t+1} \rightarrow [1.2 \ 1.3 \ 1.5]$ $x_{t+1} \rightarrow [0 \ 1 \ 0]$

6

